

Applied Algebra

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1 The Binomial Theorem

Theorem 1 (Binomial Theorem). *The Binomial theorem describes the algebraic expansion of powers of a binomial.*

$$(x + y)^n = \binom{n}{0}x^ny^0 + \binom{n}{1}x^{n-1}y^1 + \dots + \binom{n}{n}x^0y^n. \quad (1)$$

2 Problem set.

2.1 Algebra 101.

2.1.1 Problem 1.1.

Expand $(a \pm b)^5$.

$$(a \pm b)^5 \quad (2)$$

Using the Binomial Theorem (Theorem 1) yields:

$$(a \pm b)^5 = \binom{5}{0}a^5 \pm \binom{5}{1}a^4b + \binom{5}{2}a^3b^2 \pm \binom{5}{3}a^2b^3 + \binom{5}{4}ab^4 \pm \binom{5}{5}b^5. \quad (3)$$

2.1.2 Problem 1.2.

Expand $(x \pm 1)^5$.

$$(x \pm 1)^5 \quad (4)$$

Using the Binomial Theorem (Theorem 1) yields:

$$(x \pm 1)^5 = \binom{5}{0}x^5 \pm \binom{5}{1}x^4 + \binom{5}{2}x^3 \pm \binom{5}{3}x^2 + \binom{5}{4}x \pm \binom{5}{5}. \quad (5)$$

2.1.3 Problem 1.3.

Expand $(a + b + c + d)^2$.

$$(a + b + c + d)^2 = a^2 + b^2 + c^2 + d^2 + 2ab + 2ac + 2ad + 2bc + 2bd + 2cd. \quad (6)$$

2.1.4 Problem 1.4.

Rewrite $a^2 + ab + a + b$ as a product of two first-degree polynomials.

$$a^2 + ab + a + b = a(a + b) + (a + b) = (a + b)(a + 1). \quad (7)$$

2.1.5 Problem 1.5.

Rewrite $x^3 + 2x^2 + 2x + 1$ as a product of a second and a first degree polynomial. (It cannot be rewritten as a product of three first-degree polynomials, due to The Fundamental Theorem of Algebra and results from Abstract Algebra / Field Theory).

$$\begin{aligned}x^3 + 2x^2 + 2x + 1 &= (x^3 + x^2 + x) + (x^2 + x + 1) = x(x^2 + x + 1) + (x^2 + x + 1) = (8) \\ &= (x^2 + x + 1)(x + 1). \quad (9)\end{aligned}$$

2.1.6 Problem 1.6.

Rewrite $x^3 + 6x^2 + 12x + 8$ as a product of polynomials.

Method 1.

Notice that the coefficients 1, 6, 12, 8 match the binomial expansion pattern of a perfect cube. Then use the identity

$$a^3 + 3a^2b + 3ab^2 + b^3 = (a + b)^3. \quad (10)$$

Thus,

$$x^3 + 6x^2 + 12x + 8 = (x + 2)^3. \quad (11)$$

Method 2.

We observe that -2 is a root of our polynomial (i.e. a point at which the expression evaluates to 0), and thus $(x + 2)$ is a factor. Dividing the polynomial by $(x + 2)$ gives

$$x^2 + 4x + 4 = (x + 2)^2, \quad (12)$$

Therefore

$$x^3 + 6x^2 + 12x + 8 = (x + 2)(x + 2)^2 = (x + 2)^3. \quad (13)$$

2.1.7 Problem 1.7.

Find all solutions of the equation $ab + 2a - 3b = 14$ where $a, b \in \mathbb{N}^+$.

Group terms containing a

$$ab + 2a - 3b = 14$$

$$a(b + 2) - 3b = 14$$

Move the $3b$ term

$$a(b + 2) = 3b + 14$$

Express a :

$$a = \frac{3b + 14}{b + 2}$$

Rewrite the numerator:

$$3b + 14 = 3(b + 2) + 8$$

Thus

$$a = 3 + \frac{8}{b+2}.$$

For a to be integer, $b+2$ must divide 8.

Divisors of 8: 1, 2, 4, 8. Since $b \geq 1$, $b+2 \geq 3$, we are left with two possibilities:

- $b+2 = 8 \implies b = 6$
- $b+2 = 4 \implies b = 2$

Now we plug these b values back into our expression for a to find the two solutions to our equation: $(5, 2)$ and $(4, 6)$.

2.1.8 Problem 1.8.

What is the minimal point of the expression $2x^2 + 2y^2 - 2xy + 4x - 6y + 8$?

2.1.9 Problem 1.9.

Prove the following inequality:

$$(a+b+c)\left(\frac{1}{a} + \frac{1}{b} + \frac{1}{c}\right) \geq 9$$

where $a, b, c > 0$.

2.1.10 Problem 2.1.

Expand $(2x-3)^3$.

2.1.11 Problem 2.2.

Expand $(a-1)(b-1)(c-1)$.

2.1.12 Problem 2.3.

Rewrite the following expression as a product:

$$(a+b)^2 - (a-b)^2. \tag{14}$$

2.1.13 Problem 2.4.

Rewrite the following expression as a product:

$$x^2y + xy^2 - x^2 + y^2. \tag{15}$$

2.1.14 Problem 2.5.

Rewrite $a^4 - 1$ as a product.

Using difference of squares yields:

2.1.15 Problem 2.6.

Find all solutions of the equation

$$4ab - 6a + 8b = 14. \tag{16}$$

2.1.16 Problem 2.7.

Find the minimal point of the following expression

$$5x^2 + 2y^2 - 2x + 4y + 4xy + 1.$$

2.1.17 Problem 2.8.

Prove the following inequality:

$$4x + \frac{1}{x} \geq 4$$

where $x > 0$.